

Taner Giray Sönmez

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EDUCATION

- Columbia University** New York City, USA
MS in Computer Science, Machine Learning Track; GPA: 3.96 *September 2024 - December 2025*
Related Coursework: High-Performance Machine Learning, Machine Learning Cloud, Neural Networks and Deep Learning, Applied Machine Learning, Natural Language Processing, Artificial Intelligence
- Sabanci University** Istanbul, Turkey
BS in Computer Science; GPA: 3.82 *September 2019 - June 2024*
- Uppsala University** Uppsala, Sweden
BS in Computer Science; Erasmus Exchange Program *January 2023 - June 2023*

EXPERIENCE

- Amazon Web Services (AWS)** New York City, USA
Software Development Engineer Intern, AWS Glue *May 2025 - August 2025*
 - : Enhanced Personally Identifiable Information (PII) name detection in AWS Glue's Apache Spark environment by deploying a new ML model, reducing false negatives by 37% and false positives by 20%.
 - : Accelerated PII detection model inference speed by approximately 2x, improving overall processing efficiency.
 - : Expanded PII detection to include 15+ new entity types, such as organizations and locations.
 - : Evaluated 12+ machine learning models and implemented the one most suitable for AWS's setup in terms of speed and computing efficiency.
- Columbia University - Internet Real Time Lab** New York City, USA
AI Researcher *January 2025 - May 2025*
 - : Created and annotated a dataset of 450+ privacy policies covering AI training, cross-border transfers, and advertiser data sharing for LLM-based policy analysis.
 - : Benchmarked models (GPT-4o, DeepSeek, Mistral, BERT family) under zero-shot, few-shot, and Retrieval-Augmented Generation (RAG) setups, achieving up to 99% recall in clause classification.
- Mars Athletic Club (MACFIT)** Istanbul, Turkey
Data Science Intern *July 2022 - January 2023 and July 2023 - June 2024*
 - : Built an ML-based churn prediction model with XGBoost on millions of customer records, increasing contract renewals and generating \$45K+ additional monthly revenue.
 - : Conducted clustering with unsupervised learning to segment customers and improve churn prediction accuracy.
 - : Queried and preprocessed large datasets using SQL for modeling and analysis.
 - : Performed sentiment analysis with BERT on customer comments to identify key satisfaction trends and improvement areas.

PROJECTS

- LightGPT2: Efficient GPT-2 Optimization** *Jan 2025 - May 2025*
Machine Learning Engineer
 - : Built a compressed variant of the GPT-2 LLM using knowledge distillation, LoRA, 80% pruning, and 4-bit quantization.
 - : Reduced training memory by 66% and accelerated inference 3x with FlashAttention, deployable on consumer GPUs with 6GB VRAM.
- Neural Document Search and Q&A System** *Jan 2025 - May 2025*
Machine Learning Engineer
 - : Developed and fine-tuned a dual-component DistilBERT-based retriever-reader system for neural search and Q&A over document corpora.
 - : Retriever achieved Recall@20 of 74.4%, and the reader reached 86.5% F1 on SQuAD v1.1 with a lightweight architecture.
 - : Designed for scalable deployment on Google Cloud Platform.
- Sarcasm Detection in Text** *Jan 2025 - May 2025*
Machine Learning Researcher
 - : Fine-tuned and compared transformer models (RoBERTa, DeBERTa) with hierarchical RNN architectures, demonstrating transformers' superior performance on diverse sarcasm datasets.
 - : Achieved 76% accuracy on the BigBench SNARKS benchmark with a fine-tuned DeBERTa model, demonstrating strong cross-domain generalization.

SKILLS SUMMARY

- Programming Languages:** Python, C++, SQL/NoSQL
- Tools and Frameworks:** Apache Spark, CUDA, Scikit-learn, Numpy, Google Cloud Platform, Pandas, Selenium, Large Scale Data Processing, Flutter, Git, PostgreSQL, MySQL, PyTorch, MongoDB, Pinecone
- Soft Skills:** Agile Methodologies, Analytical Thinking, Teamwork, Problem Solving, Effective Communication, Team Player, Adaptability, Project Management